The Fetal Acoustic Environment may be too Noisy for Prenatal Phonetic Learning from External Sources

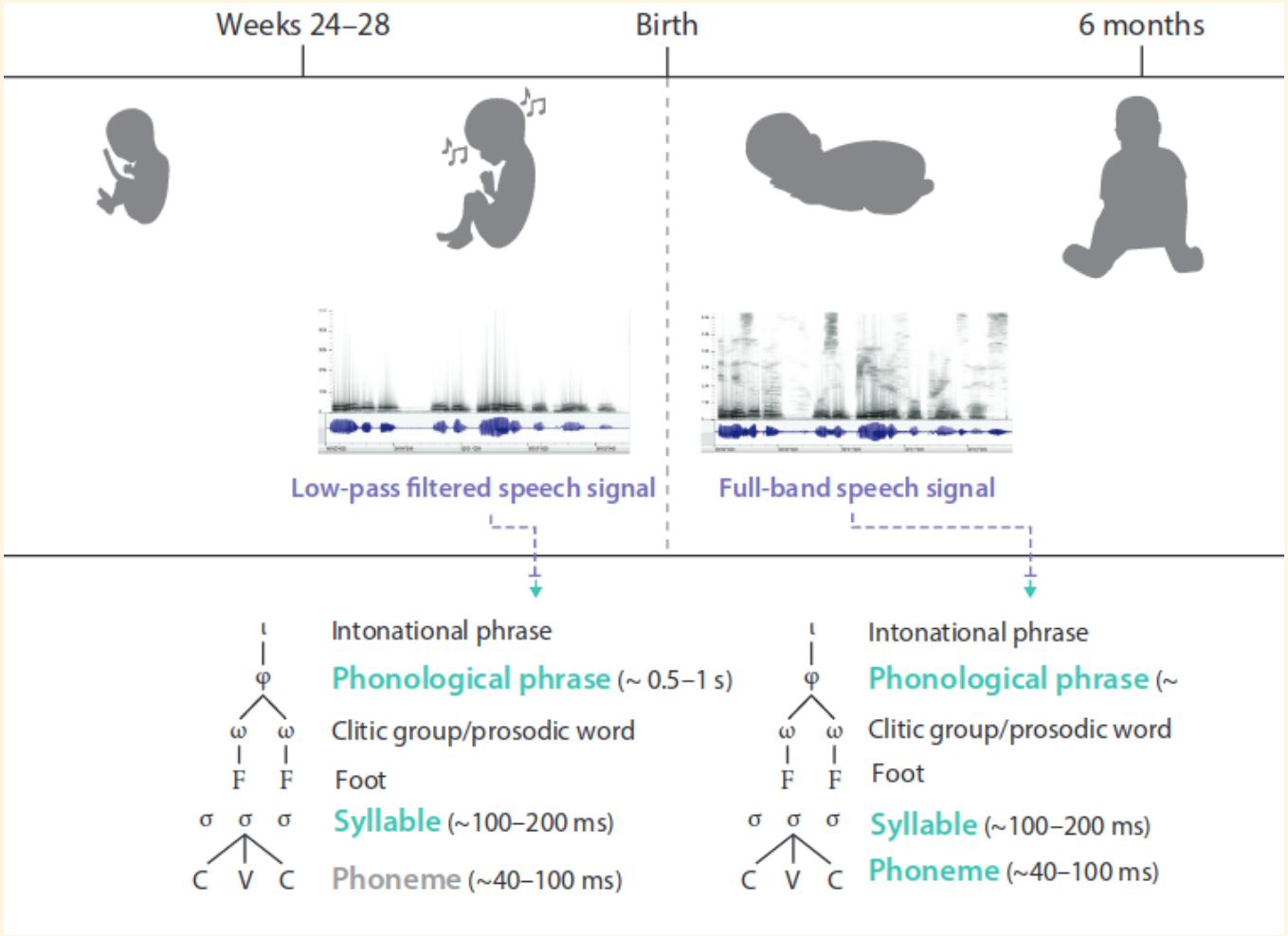
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Introduction: Sensory life starts before birth



[Nallet and Gervain, 2021]

What about phonetic learning? Evidence it could start prenatally [Moon et al., 2013, Griffiths et al., 1994] but they are sparse.

Why would prenatal phonetic learning matter? How could we study it?

- Could help to explain why infants learn vowels before consonants
- May have an adaptive value [Vogelsang et al., 2024]
- Understanding what is learnt at each stage is useful for understanding typical language acquisition and supporting infants with atypical experiences.
- But it is very challenging, if not impossible to study empirically

> ***Current study***
> Modeling study of how the prenatal acoustic environment shapes early phonetic learning

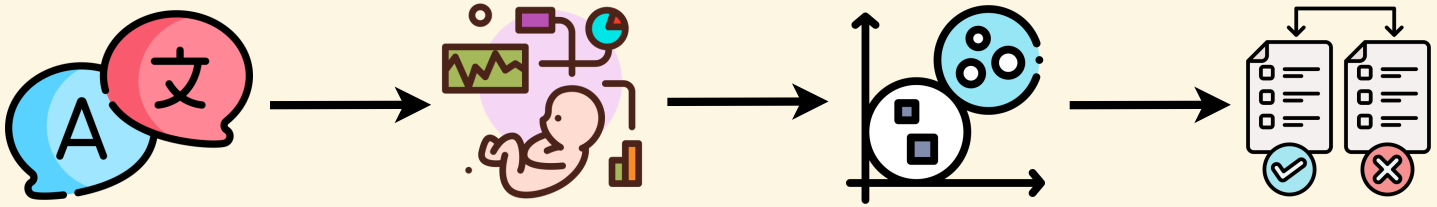
Methods: General Setup

Design a learning environment : speech dataset in a "native language"

Alter the training data so they sound like what a fetus would hear

Train a machine learning model that implements a plausible learning mechanism

Assess model's perceptual abilities compared with infants' empirical records and make novel predictions

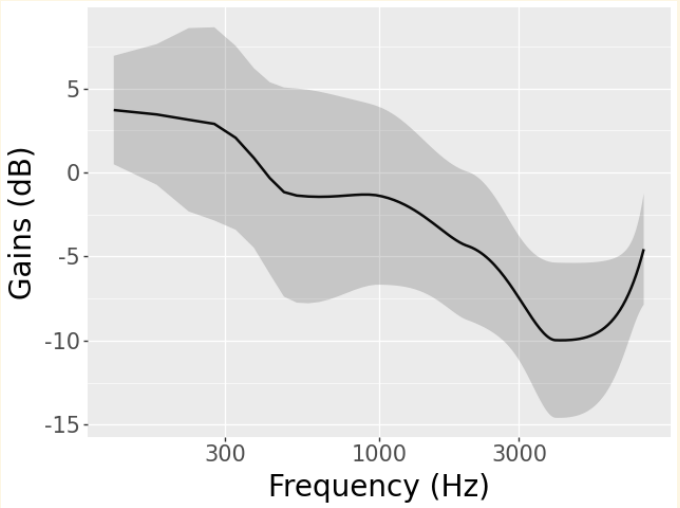


Designing the prenatal environment

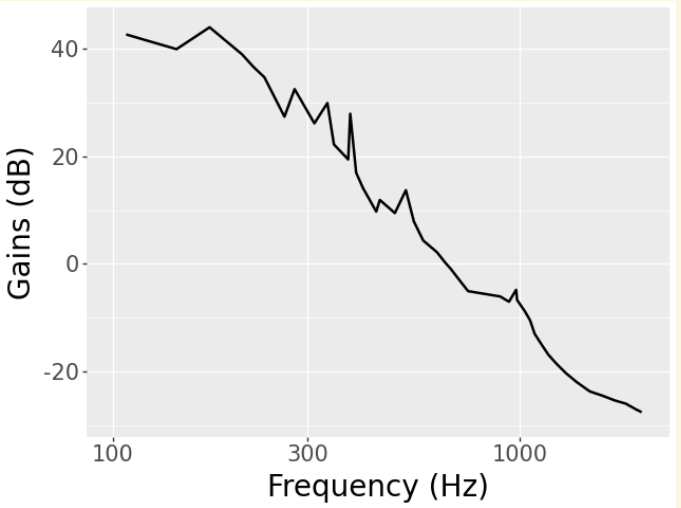
Speech datasets

Dataset	Duration	Language	Number of speakers	Reference
Wall Street Journal	19h30	American English	96	[Paul and Baker, 1992]
Globalphone	19h33	Japanese	96	[Schultz, 2002]

Prenatal acoustic environment



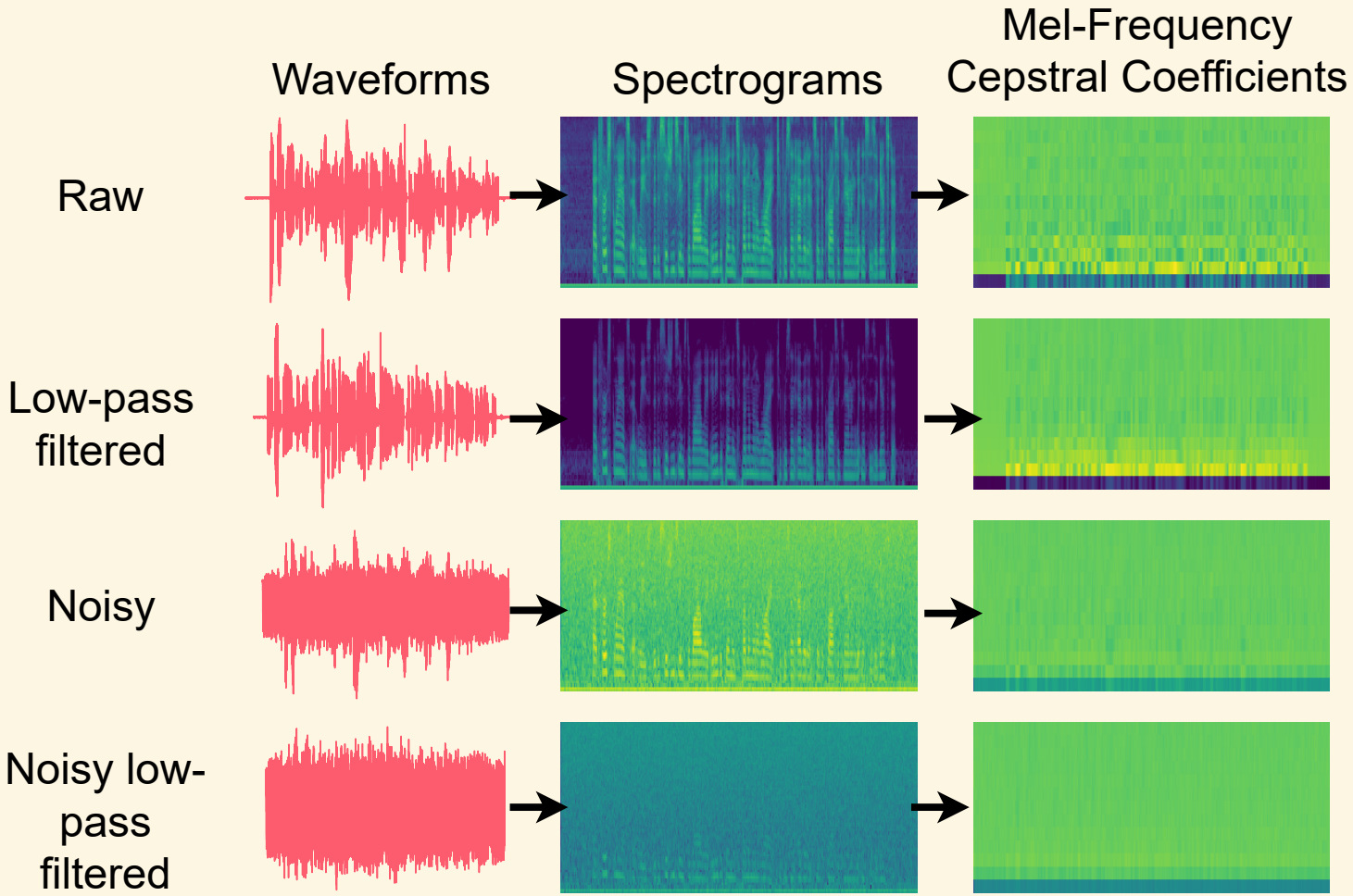
Frequency response of body filter [Richards et al., 1992]



Average background noise level [Parga et al., 2018]

Speech representations

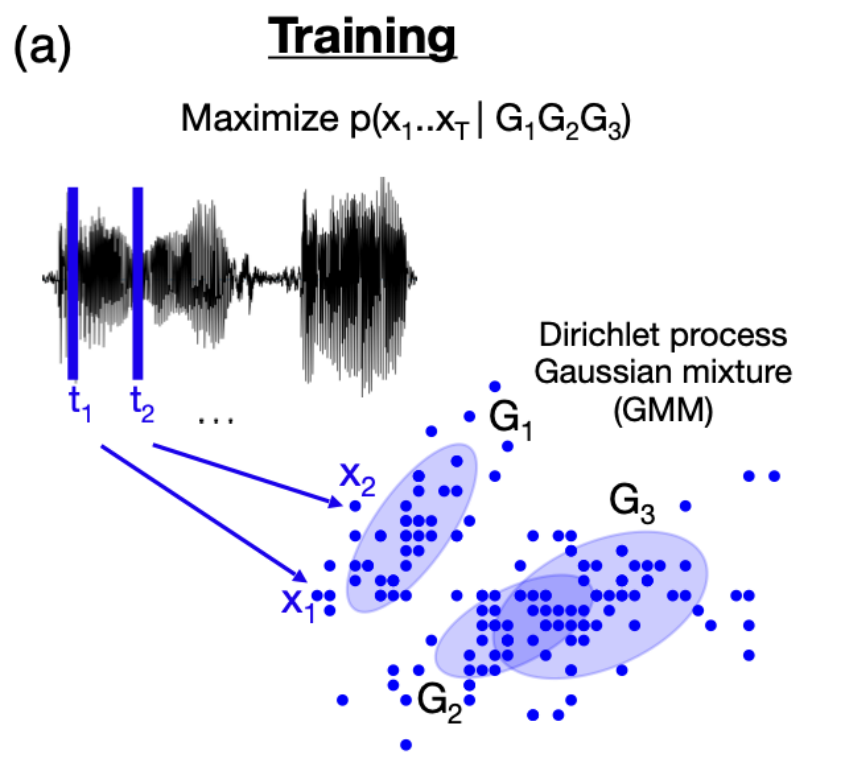




[Zahorian and Jagharghi, 1993, Iskarous and Vietti, 2025]

Simulating phonetic learning: Mechanism

Dirichlet Process Gaussian mixture models with an efficient sampling algorithm [Chang and Fisher, 2013], a plausible unsupervised model of early phonetic learning [Schatz et al., 2021]



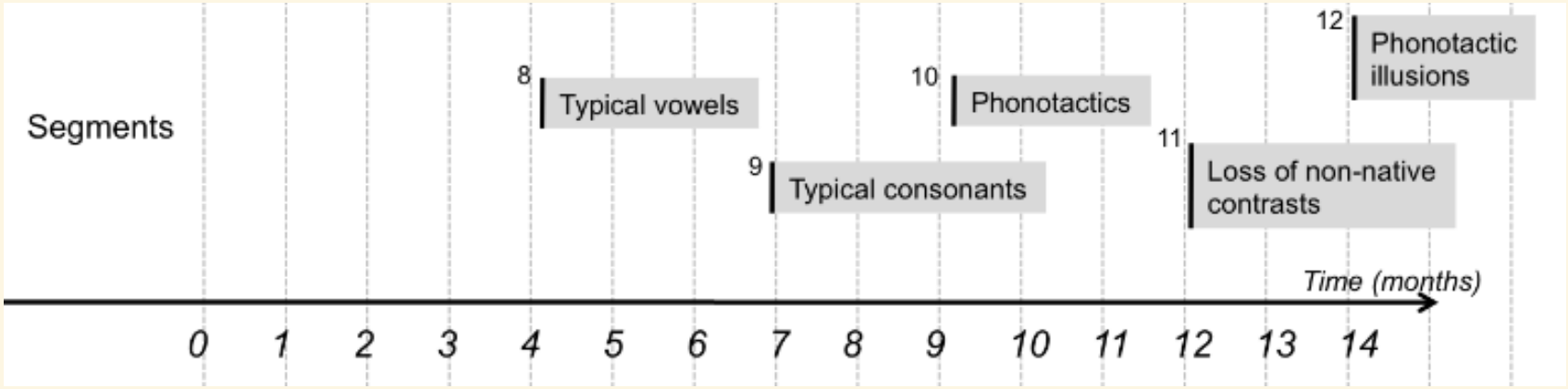
Learning outcome assessment

Machine ABX tasks [Schatz et al., 2013]

Contrast	Type	L1	L2	Reference
/ɹ/-/l/	Consonants	American English	Japanese	[Kuhl et al., 2006]
/ɚ/-/ɪ/, /ɚ/-/ʌ/	Vowels	American English	Japanese	[Schatz et al., 2021]

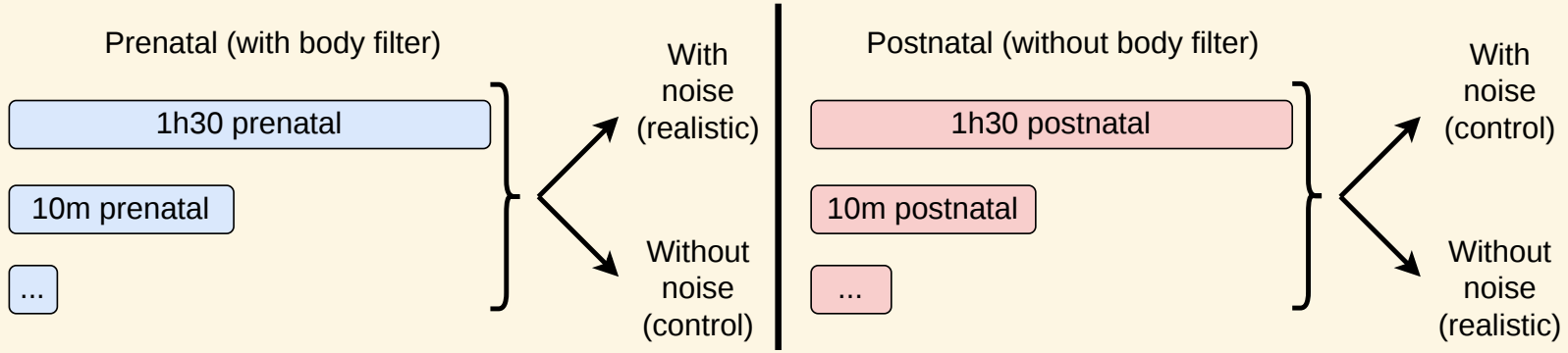


Experiment 1: Effect on contrasts learning order?

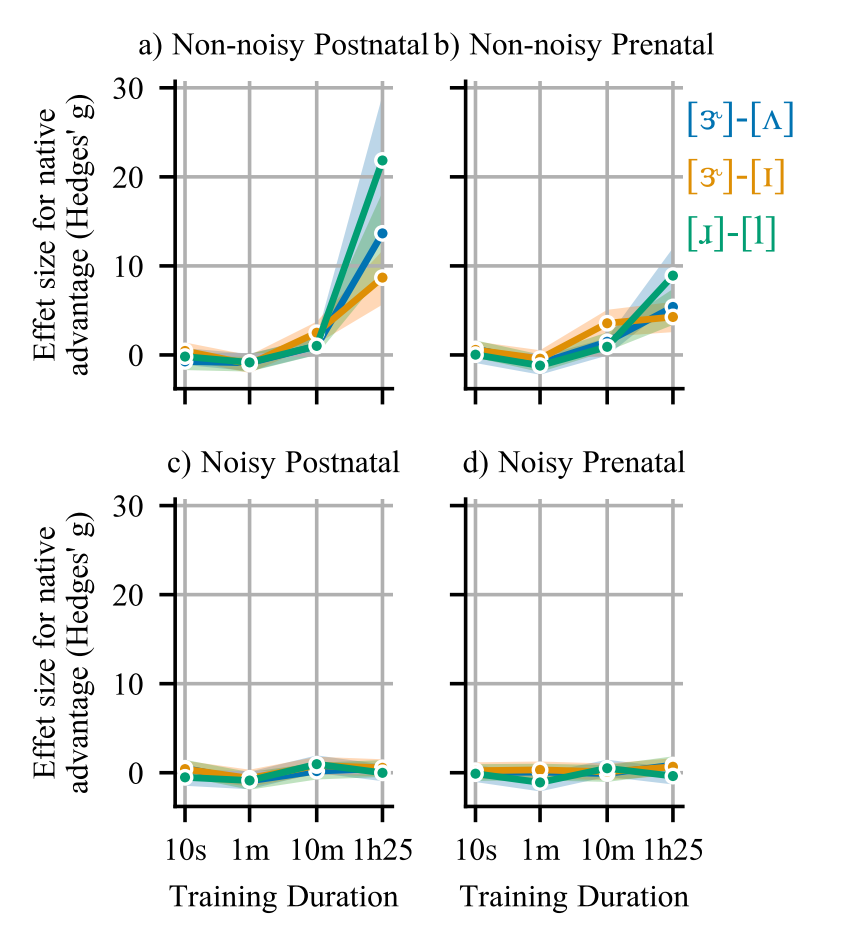


[Dupoux, 2018]

> ***Question***
>
> Could the sequence in which contrasts are learned be influenced by the prenatal acoustic environment?



Experiment 1: Results

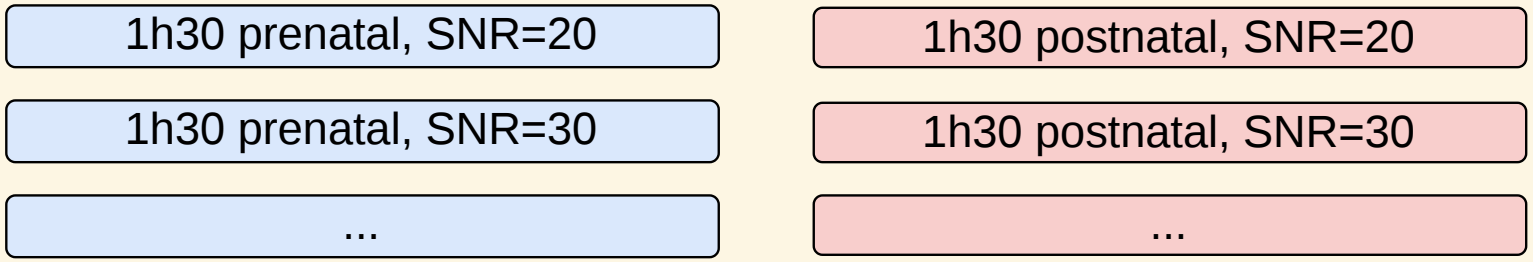


- There is no statistically significant difference in the timing of vowel and consonant acquisition.
- Models struggle to learn any native advantage in realistic conditions.

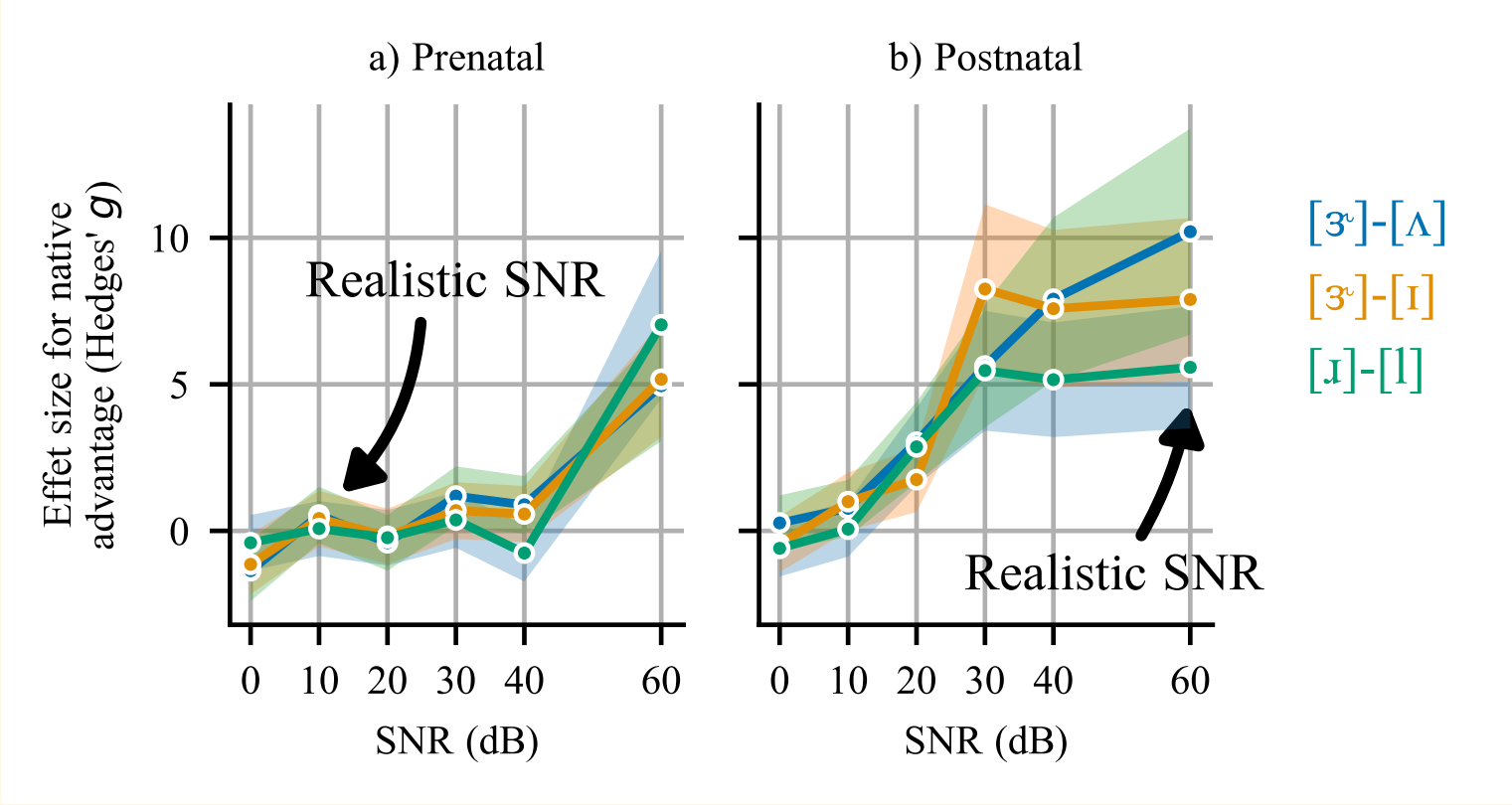
Experiment 2: Is prenatal phonetic learning even possible?

Was the signal-to-noise ratio that we used too low (10dB according to [Parga et al., 2018])?

> ***Question***
> Is there a Signal-to-Noise ratio value where prenatal phonetic learning can happen ? [Moon et al., 2013]



Experiment 2: Results



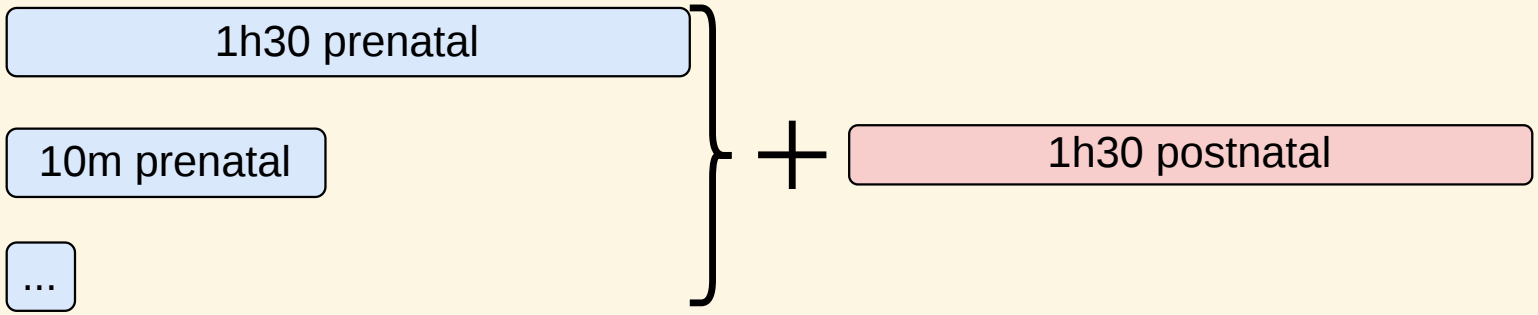
Native advantage significantly emerges at 30 dB with postnatal speech, but not before 60dB with prenatal speech.

It seems very hard to learn anything from outside sources in realistic prenatal conditions.

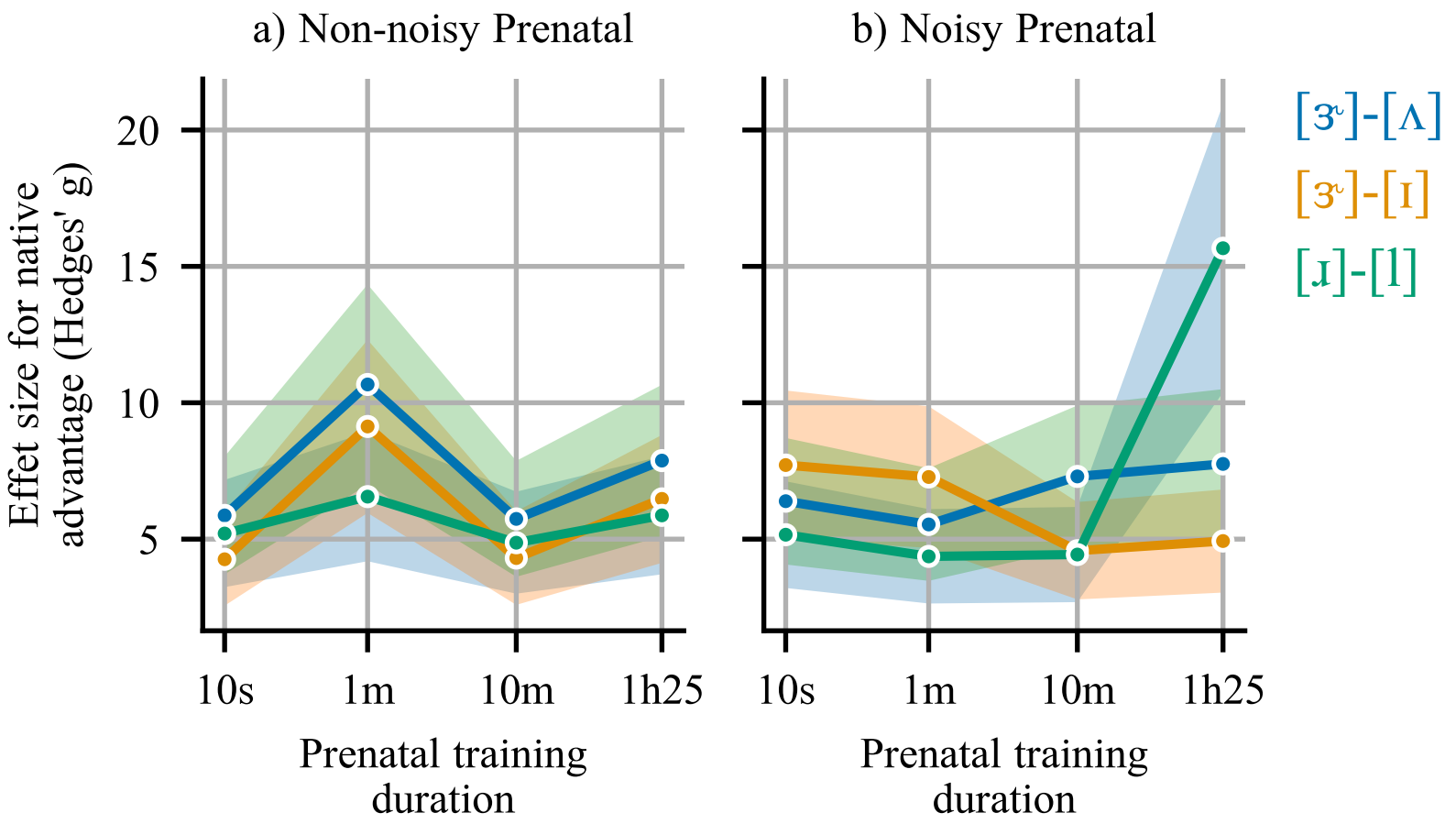
Experiment 3: Is prenatal exposure beneficial for postnatal learning?

We found no prenatal phonetic learning. However, prenatal exposure may still prove beneficial for subsequent learning [Vogelsang et al., 2024].

> ***Question***
> Do models that are exposed to more prenatal speech develop a larger native advantage?



Experiment 3: Results



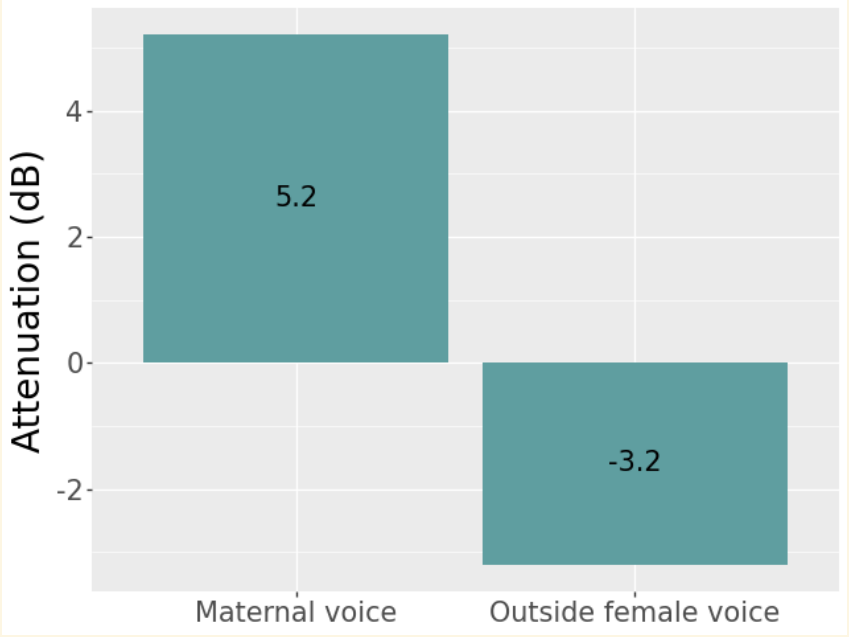
Training with degraded input is not significantly beneficial for subsequent learning.

Conclusion

- Experiment 3: No benefit of prenatal exposure for postnatal learning... But what if we had simulated ”birth” differently?
- Experiment 1: Delayed specialization for consonants may not be the consequence of prenatal speech exposure, but of other factors. There are simulated evidence for the vowel contrasts, but no empirical evidence !
- Experiment 2 & overall: Prenatal phonetic learning seems very hard in realistic condition... According to the papers we chose to design the filter

Perspectives

- What if input comes from the maternal voice?
- Simulate the noise amplitude modulations, which may make the task easier



[Richards et al., 1992]

Acknowledgements

Thank you for listening!

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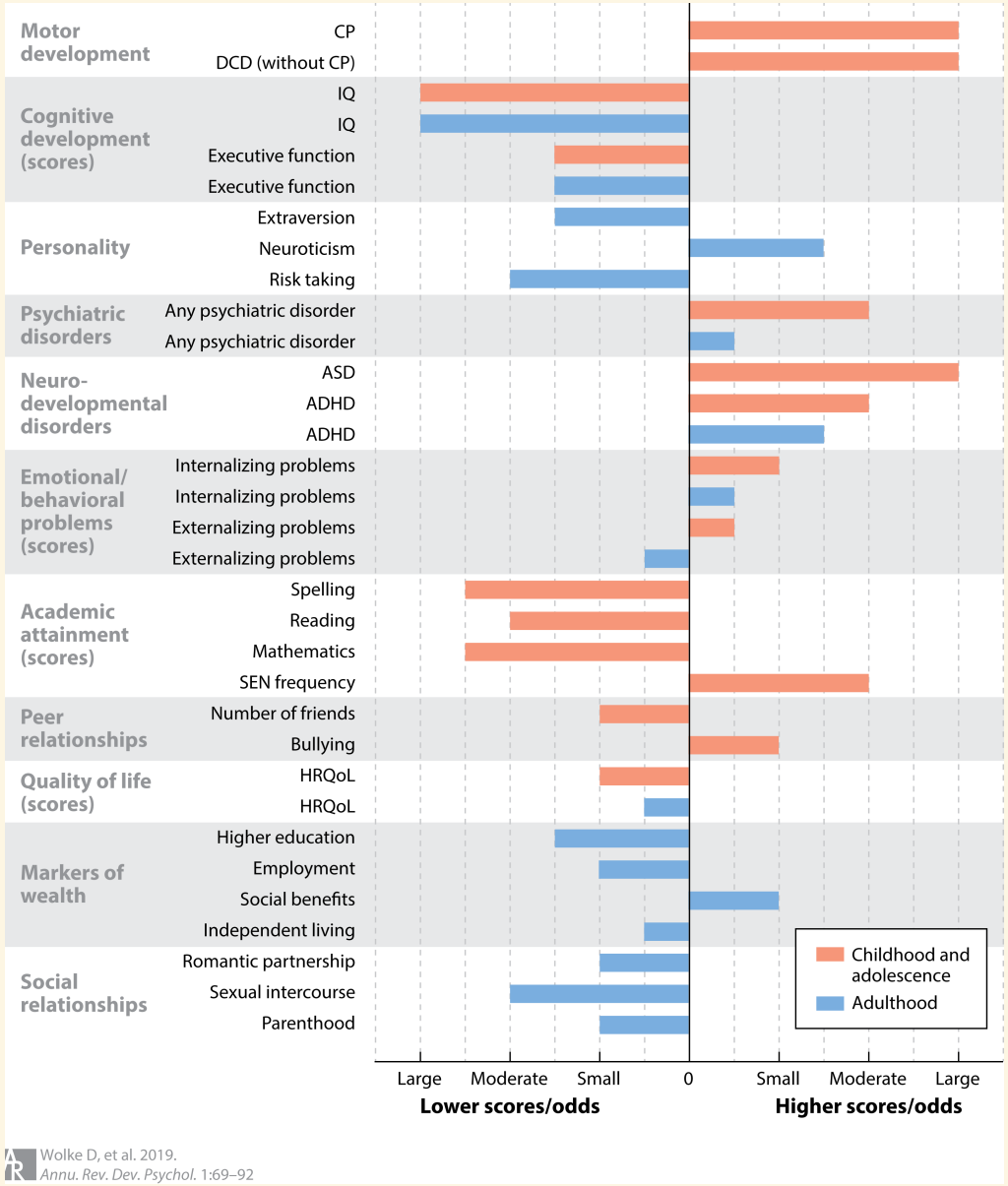
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Preterm infants and comorbidities



from [Wolke et al., 2019]

Experiment 1 figure 2

